	Result	Size	Time	Cycles	GPU	SM Frequency	Process	Attributes
Current	680 - flash_attention	(512, 1, 1)x(128, 1, 1)	12.25 ms	17,071,075	0 - NVIDIA GeForce RTX 3090	1.39 Ghz	[124698] python3.12	

GPU Speed Of Light Throughput

All

High-level overview of the throughput for compute and memory resources of the GPU. For each unit, the throughput reports the achieved percentage of utilization with respect to the theoretical maximum. Breakdowns show the throughput for each individual sub-metric of Compute and Memory to clearly identify the highest contributor. High-level overview of the utilization for compute and memory resources of the GPU presented as a roofline chart.

Compute (SM) Throughput [%]	38.40	Duration [ms]	12.25
Memory Throughput [%]	33.37	Elapsed Cycles [cycle]	17071075
L1/TEX Cache Throughput [%]	26.10	SM Active Cycles [cycle]	15243857.66
L2 Cache Throughput [%]	33.37	SM Frequency [Ghz]	1.39
DRAM Throughput [%]	1.16	DRAM Frequency [Ghz]	9.49

Latency Issue This workload exhibits low compute throughput and memory bandwidth utilization relative to the peak performance of this device. Achieved compute throughput and/or memory bandwidth below 60.0% of peak typically indicate latency issues. Look at Scheduler Statistics and Warp State Statistics for potential reasons.

Key Performance Indicators

GPU Throughput

Compute Throughput Breakdown

SM: Pipe Tensor Cycles Active [%]	38.40
SM: Issue Active [%]	25.96
SM: Inst Executed [%]	25.96
SM: Inst Executed Pipe Lsu [%]	19.34
SM: Pipe Fmaheavy Cycles Active [%]	14.55
SM: Pipe Alu Cycles Active [%]	13.67
SM: Pipe Fma Cycles Active [%]	13.27
SM: Mio2rf Writeback Active [%]	10.86
SM: Mio Inst Issued [%]	6.87
SM: Mio Pq Write Cycles Active [%]	2.43
SM: Mio Pq Read Cycles Active [%]	2.43
SM: Inst Executed Pipe Adu [%]	1.29
SM: Inst Executed Pipe Cbu Pred On Any [%]	0.60
SM: Inst Executed Pipe Uniform [%]	0.34
SM: Inst Executed Pipe Xu [%]	0.00
SM: Memory Throughput Internal Activity [%]	0
IDC: Request Cycles Active [%]	0
SM: Instruction Throughput Internal Activity [%]	0
SM: Inst Executed Pipe Tex [%]	0
SM: Pipe Fp64 Cycles Active [%]	0
SM: Inst Executed Pipe Ipa [%]	0

Memory Throughput Breakdown

L2: T Sectors [%]	33.37
L2: Lts2xbar Cycles Active [%]	33.13
L1: Data Pipe Lsu Wavefronts [%]	23.34
L1: Lsu Writeback Active [%]	21.67
L1: Lsuin Requests [%]	19.34
L1: M Xbar2l1tex Read Sectors [%]	19.22
L1: Data Bank Reads [%]	9.61
L2: D Sectors [%]	8.41
L2: T Tag Requests [%]	8.34
GPU: Compute Memory Access Throughput Internal Activity [%]	8.30
L2: Xbar2lts Cycles Active [%]	4.18
L1: M L1tex2xbar Req Cycles Active [%]	2.42
L1: Data Bank Writes [%]	2.40
DRAM: Cycles Active [%]	1.16
DRAM: Dram Sectors [%]	1.00
L2: D Sectors Fill Device [%]	0.46
L1: Texin Sm2tex Req Cycles Active [%]	0.15
L2: D Sectors Fill Sysmem [%]	0.00
L1: Tex Writeback Active [%]	0
L2: D Atomic Input Cycles Active [%]	0
L1: F Wavefronts [%]	0
L1: Data Pipe Tex Wavefronts [%]	0
GPU: Compute Memory Request Throughput Internal Activity [%]	0

Floating Point Operations Roofline

Performance [FLOP/s] (1 = 1e+12)
Arithmetic Intensity [FLOP/byte]

Floating Point Operations Roofline (Double Precision)

Performance [FLOP/s] (1 = 1e+12)
Arithmetic Intensity [FLOP/byte]

Floating Point Operations Roofline (Half Precision)

Performance [FLOP/s] (1 = 1e+13)
Arithmetic Intensity [FLOP/byte]

Floating Point Operations Roofline (Single Precision)

Performance [FLOP/s] (1 = 1e+13)
Arithmetic Intensity [FLOP/byte]

Tensor Core Operations Roofline

Performance [OP/s] (1 = 1e+14)
Arithmetic Intensity [OP/byte]

	# Operations	# Operations / Cycle	# Operations / s	Peak %	eak Operations / Cycle	Peak Operations / s
Src:bf16 Dst:fp32 Sparsity:	549755813888	32246.07	44889106820165.23	76.81	41984	58445081255650.39
Src:bf16 Dst:fp32 Sparsity:	0	0	0	0	83968	116890162511300.78
Src:fp16 Dst:fp16 Sparsity:	0	0	0	0	83968	116890162511300.78
Src:fp16 Dst:fp16 Sparsity:	0	0	0	0	167936	233780325022601.56
Src:fp16 Dst:fp32 Sparsity:	0	0	0	0	41984	58445081255650.39
Src:fp16 Dst:fp32 Sparsity:	0	0	0	0	83968	116890162511300.78
Src:fp64	0	0	0	0	289.54	403069525901.04
Src:int1	0	0	0	0	1343488	1870242600180812...
Src:int4 Sparsity:off	0	0	0	0	335872	467560650045203.12
Src:int4 Sparsity:on	0	0	0	0	671744	935121300090406.25
Src:int8 Sparsity:off	0	0	0	0	167936	233780325022601.56
Src:int8 Sparsity:on	0	0	0	0	335872	467560650045203.12
Src:tf32 Dst:fp32 Sparsity:c	0	0	0	0	20992	29222540627825.20
Src:tf32 Dst:fp32 Sparsity:c	0	0	0	0	41984	58445081255650.39

PM Sampling

Timeline view of PM metrics sampled periodically over the workload duration. Data is collected across multiple passes. Use this section to understand how workload behavior changes over its runtime.

Maximum Sampling Interval [us]	4	# Pass Groups	2
Maximum Buffer Size [Mbyte]	32.83	-	-

Compute Workload Analysis

All

Detailed analysis of the compute resources of the streaming multiprocessors (SM), including the achieved instructions per clock (IPC) and the utilization of each available pipeline. Pipelines with very high utilization might limit the overall performance.

Executed Ipc Elapsed [inst/cycle]	1.04	SM Busy [%]	38.40
Executed Ipc Active [inst/cycle]	1.16	Issue Slots Busy [%]	25.96
Issued Ipc Active [inst/cycle]	1.16		

Pipe Utilization (% of elapsed cycles)

Pipe Utilization (% of peak instructions executed over elapsed cycles)

Pipe Utilization (% of active cycles)

Pipe Utilization (% of peak instructions executed over active cycles)

Memory Workload Analysis

All

Detailed analysis of the memory resources of the GPU. Memory can become a limiting factor for the overall kernel performance when fully utilizing the involved hardware units (Mem Busy), exhausting the available communication bandwidth between those units (Max Bandwidth), or by reaching the maximum throughput of issuing memory instructions (Mem Pipes Busy). Detailed chart of the memory units. Detailed tables with data for each memory unit.

Memory Throughput [Gbyte/s]	10.57	Mem Busy [%]	33.37
L1/TEX Hit Rate [%]	0	Max Bandwidth [%]	33.13
L2 Hit Rate [%]	98.62	Mem Pipes Busy [%]	19.34
L2 Compression Input Sectors [sector]	0	Local Memory Spilling Requests	0
L2 Compression Ratio	0	Local Memory Spilling Request Overhead [%]	0
L2 Compression Success Rate [%]	0	L2 Persisting Size [Mbyte]	1.18

Memory Chart

Values: Throughput Inactivity: Greyed Out

Shared Memory

	Instructions	Requests	Wavefronts	% Peak	Bank Conflicts
Shared Load	0	0	268594308	19.21	12268
Shared Load Matrix	67108864	67108864			
Shared Store	0	0	0	0	0
Shared Store From Global Lc	16777216	16777216	67108864	1.20	0
Shared Atomic	0	0	0	0	0
Other	-	-	49163347	3.52	0
Total	83886080	83886080	384866519	23.93	12268

L1/TEX Cache

	Instructions	Requests	Wavefronts	% Peak	Sectors	Sectors/Req	Hit Rate	Bytes	Sector
Local Load	0	0	0	0	0	0	0	0	
Global Load	16384	16384	33589190	2.40	262144	16	0	8388608	
Global Load To Shared Store (acce	0	0			0	0		0	
Global Load To Shared Store (bypa	16777216	16777216			268435456	16	-	8589934592	
Surface Load	0	0	0	0	0	0	0	0	
Texture Load	0	0	0	0	0	0	0	0	
Global Store	16384	16384	32929	0.00	262144	16	0	8388608	
Local Store	0	0	0	0	0	0	0	0	
Surface Store	0	0	0	0	0	0	0	0	
Global Reduction	0	0	0	0	0	0	0	0	
Surface Reduction	0	0	0	0	0	0	0	0	
Global Atomic ALU	0	0	0	0	0	0	0	0	
Global Atomic CAS	0								
Surface Atomic ALU	0	0	0	0	0	0	0	0	
Surface Atomic CAS	0								
Loads	16793600	16793600	33589190	2.40	268697600	16	0	8598323200	
Stores	16384	16384	32929	0.00	262144	16	0	8388608	
Atomics & Reductions	0	0	0	0	0	0	0	0	

L2 Cache

	ors	Sectors/Req	% Peak	Hit Rate	Bytes	Throughput	Sector Misses to Devic	ector Misses to Systen	Sector Misses to Peer
L1/TEX Load	70436576	4.00	33.34	98.62	8653970432	706621000318.77	3710088	0	0
L1/TEX Store	262144	4	0.03	100	8388608	684953412.17	0	0	0
L1/TEX Atomic ALU	0	0	0	0	0	0	0	0	0
L1/TEX Atomic CAS	0	0	0	0	0	0	0	0	0
L1/TEX Reduction	0	0	0	0	0	0	0	0	0
L1/TEX Total	70692872	4.00	33.37	98.62	8662171904	707290673550.76	3728584	0	0
GCC Total	13328	4	0.00	85.80	426496	34824596.70	1892	0	0
ECC Total	0	-	0	-	0	0	0	-	-

L2 Cache Eviction Policies

	First	Hit Rate	Last	Hit Rate	Normal	Hit Rate	Normal Demote	Hit Rate
L1/TEX Load	0	0	270178560	98.71	262144	0	0	0
L1/TEX Store	0	0	0	0	262144	100	0	0
L1/TEX Atomic	0	0	0	0	0	0	-	-
L1/TEX Total	0	0	270154420	98.72	524288	50	0	0
GPU Total	10234	55.35	270158108	98.72	541086	51.20	0	0

Device Memory

	Sectors	% Peak	Bytes	Throughput
Load	3728140	1.07	119300480	9741219383.46
Store	317120	0.09	10147840	828599647.78
Total	4045260	1.16	129448320	10569819031.24

Scheduler Statistics

Summary of the activity of the schedulers issuing instructions. Each scheduler maintains a pool of warps that it can issue instructions for. The upper bound of warps in the pool (Theoretical Warps) is limited by the launch configuration. On every cycle each scheduler checks the state of the allocated warps in the pool (Active Warps). Active warps that are not stalled (Eligible Warps) are ready to issue their next instruction. From the set of eligible warps the scheduler selects a single warp from which to issue one or more instructions (Issued Warp). On cycles with no eligible warps, the issue slot is skipped and no instruction is issued. Having many skipped issue slots indicates poor latency hiding.

Active Warps Per Scheduler [warp]	2.65	No Eligible [%]	70.96
Eligible Warps Per Scheduler [warp]	0.42	One or More Eligible [%]	29.04
Issued Warp Per Scheduler	0.29		

Warps Per Scheduler

GPU Maximum Warps Per Scheduler
Theoretical Warps Per Scheduler
Active Warps Per Scheduler
Eligible Warps Per Scheduler
Issued Warp Per Scheduler
0.0
4.0
8.0
12.0
16.0

Warp State Statistics

Analysis of the states in which all warps spent cycles during the kernel execution. The warp states describe a warp's readiness or inability to issue its next instruction. The warp cycles per instruction define the latency between two consecutive instructions. The higher the value, the more warp parallelism is required to hide this latency. For each warp state, the chart shows the average number of cycles spent in that state per issued instruction. Stalls are not always impacting the overall performance nor are they completely avoidable. Only focus on stall reasons if the schedulers fail to issue every cycle. When executing a kernel with mixed library and user code, these metrics show the combined values.

Warp Cycles Per Issued Instruction [cycle]	9.12	Avg. Active Threads Per Warp	32
Warp Cycles Per Executed Instruction [cycle]	9.12	Avg. Not Predicated Off Threads Per Warp	31.02

Warp State (All Cycles)

Instruction Statistics

All

Statistics of the executed low-level assembly instructions (SASS). The instruction mix provides insight into the types and frequency of the executed instructions. A narrow mix of instruction types implies a dependency on few instruction pipelines, while others remain unused. Using multiple pipelines allows hiding latencies and enables parallel execution. Note that 'Instructions/Opcode' and 'Executed Instructions' are measured differently and can diverge if cycles are spent in system calls.

Executed Instructions [inst]	1451810816	Avg. Executed Instructions Per Scheduler [inst]	4426252.49
Issued Instructions [inst]	1451819490	Avg. Issued Instructions Per Scheduler [inst]	4426278.93

Executed Instruction Categories

Executed Instruction Mix

NVLink Topology

NVLink Topology diagram shows logical NVLink connections with transmit/receive throughput.

NVLink Topology

The system does not have any NVLink connections.

NVLink Tables

Detailed tables with properties for each NVLink.

Logical NVLink Properties

The system does not have any NVLink connections.

NUMA Affinity

Non-uniform memory access (NUMA) affinities based on compute and memory distances for all GPUs.

NUMA ID Table

GPU ID	GPU Name	NUMA ID by CPU Affini	CPU Affinity	NUMA ID by Memory Affinity
0	NVIDIA GeForce RTX 3090	0	0-29	0

Launch Statistics

Summary of the configuration used to launch the kernel. The launch configuration defines the size of the kernel grid, the division of the grid into blocks, and the GPU resources needed to execute the kernel. Choosing an efficient launch configuration maximizes device utilization.

Grid Size	512	Function Cache Configuration	CachePreferNone
Registers Per Thread [register/thread]	168	Static Shared Memory Per Block [Kbyte/block]	32.77
Block Size	128	Dynamic Shared Memory Per Block [byte/block]	0
Threads [thread]	65536	Driver Shared Memory Per Block [Kbyte/block]	1.02
Waves Per SM	2.08	Shared Memory Configuration Size [Kbyte]	102.40
Uses Green Context	0	Stack Size	1024
# SMs [SM]	82	# TPCs	41
Enabled TPC IDs	all	-	-

Tail Effect
Est. Speedup: 33.33% A wave of thread blocks is defined as the maximum number of blocks that can be executed in parallel on the target GPU. The number of blocks in a wave depends on the number of multiprocessors and the theoretical occupancy of the kernel. This kernel launch results in 2 full waves and a partial wave of 21 thread blocks. Under the assumption of a uniform execution duration of all thread blocks, this partial wave may account for up to 33.3% of the total runtime of this kernel. Try launching a grid with no partial wave. The overall impact of this tail effect also lessens with the number of full waves executed for a grid. See the Hardware Model description for more details on launch configurations.

Key Performance Indicators

Occupancy

All

Occupancy is the ratio of the number of active warps per multiprocessor to the maximum number of possible active warps. Another way to view occupancy is the percentage of the hardware's ability to process warps that is actively in use. Higher occupancy does not always result in higher performance, however, low occupancy always reduces the ability to hide latencies, resulting in overall performance degradation. Large discrepancies between the theoretical and the achieved occupancy during execution typically indicates highly imbalanced workloads.

Theoretical Occupancy [%]	25	Block Limit Registers [block]	3
Theoretical Active Warps per SM [warp]	12	Block Limit Shared Mem [block]	3
Achieved Occupancy [%]	22.07	Block Limit Warps [block]	12
Achieved Active Warps Per SM [warp]	10.59	Block Limit SM [block]	16

Theoretical Occupancy
Est. Local Speedup: 75.00% The 3.00 theoretical warps per scheduler this kernel can issue according to its occupancy are below the hardware maximum of 12. This kernel's theoretical occupancy (25.0%) is limited by the number of required registers, and the required amount of shared memory.

Key Performance Indicators

Impact of Varying Register Count Per Thread

Impact of Varying Block Size

Impact of Varying Shared Memory Usage Per Block

Impact of Varying Register Count Per Thread

Impact of Varying Block Size

Impact of Varying Shared Memory Usage Per Block

GPU and Memory Workload Distribution

Analysis of workload distribution in active cycles of SM, SMP, SMSP, L1 & L2 caches, and DRAM

Average SM Active Cycles [cycle]	15243857.66	Average L1 Active Cycles [cycle]	15243857.66
Average L2 Active Cycles [cycle]	11406274.12	Average SMSP Active Cycles [cycle]	15242622.30
Average DRAM Active Cycles [cycle]	1348420	Total SM Elapsed Cycles [cycle]	1397999038
Total L1 Elapsed Cycles [cycle]	1397999038	Total L2 Elapsed Cycles [cycle]	811237728
Total SMSP Elapsed Cycles [cycle]	5591996152	Total DRAM Elapsed Cycles [cycle]	1395050496

SMs Workload Imbalance
Est. Speedup: 9.45% One or more SMs have a much higher number of active cycles than the average number of active cycles. Maximum instance value is 10.57% above the average, while the minimum instance value is 4.00% below the average.

Key Performance Indicators

SMSPs Workload Imbalance
Est. Speedup: 9.49% One or more SMSPs have a much higher number of active cycles than the average number of active cycles. Maximum instance value is 10.62% above the average, while the minimum instance value is 4.05% below the average.

Key Performance Indicators

L1 Slices Workload Imbalance
Est. Speedup: 9.45% One or more L1 Slices have a much higher number of active cycles than the average number of active cycles. Maximum instance value is 10.57% above the average, while the minimum instance value is 4.00% below the average.

Key Performance Indicators

Workload Distribution

	Average	Min	Max	Sum
SM Active Cycles	15243857.66	14633959	17045672	1249996328
SMSP Active Cycle	15242622.30	14625321	17053264	4999580114
L1 Active Cycles	15243857.66	14633959	17045672	1249996328
L2 Active Cycles	11406274.12	11339256	11489244	547501158
DRAM Active Cycle	1348420	1345440	1350432	16181040

Source Counters

Source metrics, including branch efficiency and sampled warp stall reasons. Warp Stall Sampling metrics are periodically sampled over the kernel runtime. They indicate when warps were stalled and couldn't be scheduled. See the documentation for a description of all stall reasons. Only focus on stalls if the schedulers fail to issue every cycle.

Branch Instructions [inst]	30580736	Branch Efficiency [%]	100
Branch Instructions Ratio [%]	0.02	Avg. Divergent Branches [branches]	0

Warp Stall Sampling (All Samples)

Location	Value	Value (%)
0x7a4023287ae0 in flash_at...	9,415	2
0x7a4023288a20 in flash_at...	9,170	2
0x7a402328a010 in flash_at...	8,687	1
0x7a40232873a0 in flash_at...	8,460	1
0x7a40232848d0 in flash_at...	8,256	1

Most Instructions Executed

Location	Value	Value (%)
0x7a40232862c0 in flash_at...	600,064	0
0x7a40232862b0 in flash_at...	600,064	0
0x7a40232862a0 in flash_at...	600,064	0
0x7a4023285ef0 in flash_att...	600,064	0
0x7a4023285ee0 in flash_at...	600,064	0

To customize your report even further, you might want to learn about custom sections and writing your own rules.
You might also want to consider adding individual metrics.